

Aryan Kaul

202-740-0833 | aryan.kaul@columbia.edu | [linkedin.com/in/aryankaul](https://www.linkedin.com/in/aryankaul) | [akaul02.github.io](https://github.com/akaul02) | medium.com/@kaularyan33

EDUCATION

Columbia University

New York, NY

Master of Science, Computer Science, Thesis Track (GPA: 4.13)

May 2027

University of Maryland

College Park, MD

Bachelor of Science, Computer Science (GPA: 3.8)

May 2023

- Dean's List, all semesters. Minor in Business Analytics, Robert H. Smith School of Business.
- Teaching Assistant, CS 250 (Discrete Structures); led sessions for 40+ students and supported course tooling.

EXPERIENCE

Graduate Research Assistant

2026 – Present

Columbia University

New York, NY

- Built and evaluated a multi-agent vision-language model (VLM) auditor for deceptive Android flows, combining screen-sequence reasoning, UIAutomator-based inspection, benchmark design, and failure analysis across real apps.
- Built agentic software for human-AI interaction, including conversationally customizable productivity interfaces (e.g., Email) and visual C4/Git diffs that track structural code changes and architectural drift across agent turns.
- Designed and implemented cross-platform accessibility software spanning native Android and iOS integrations, browser extensions, and screen-reader plugins, iterating through evaluations with 50+ sighted and 15 blind users.

Software Engineer, Mobile Platforms

December 2023 – January 2026

Amazon

Seattle, WA

- Modernized Kindle's Android and Fire Tablet platform (1M+ LOC) using MVVM architecture with Kotlin and Jetpack Compose, increasing engagement on large-form-factor devices by 2% for over 10 million readers.
- Spearheaded app stability and build reliability by fixing 25+ crashes/freezes and triaging 200+ issues via stack-trace analysis, instrumenting client/server SLIs in CloudWatch, and adding static-analysis gates in a 50+ dev org.
- Built 10+ accessible component APIs and design tokens for Kindle's cross-platform design system, using task-specific coding models to accelerate development by an estimated 100+ engineering weeks and cut design drift by 70%.
- Championed accessibility across native and embedded web surfaces, advising on semantics, focus, and contrast. Resolved 90 tickets and reviewed 200+ PRs, mapping WCAG/ARIA requirements across implementations.

Engineering Intern, Infrastructure Solutions Group

May 2021 – Sep 2021

Dell Technologies

Hopkinton, MA

- Built automated reporting and telemetry pipelines across 700+ ISG programs using Atlassian REST APIs and JQL, replacing manual status decks with real-time dashboards for executive planning and OKR tracking.

TECHNICAL TALKS

North East AI Agents Day, Jane Street

May 2026

- Presented a multi-agent vision-language model (VLM) auditor for deceptive mobile flows, covering screen-sequence reasoning, automated UI inspection, evaluation design, and failure analysis across real apps.

Story I/O Vibe Coding Workshop, Columbia Digital Storytelling Lab

April 2026

- Co-facilitated an AI prototyping workshop, guiding artists, designers, and programmers from prompt decomposition through coding, debugging, and deployment using coding agents and AI-native development platforms.

Global Accessibility Awareness Month (GAAM) Conference, Amazon

May 2025

- Presented to 60+ attendees on modernizing Kindle's large-scale mobile platform, covering staged framework migration, accessible design-system adoption, validation guardrails, and regression prevention across mobile/web.

TECHNICAL SKILLS

Programming Languages: Python, Kotlin, Java, Swift, TypeScript, JavaScript, SQL

Machine Learning: PyTorch, scikit-learn, XGBoost, Neural Networks, Transformers, Model Training, Evaluation

AI Systems: LLMs, VLMs, RAG, Supervised Fine-Tuning, Multi-Agent Systems, Tool-Using Agents

AI Development: Claude Code, Codex, Cursor, Amazon Q Developer, Replit, Lovable, Figma Make

Mobile and Web: Android SDK, Jetpack Compose, AccessibilityService, UIAutomator, WebView, Chrome Extensions

Backend and Cloud: Supabase, Firebase, Docker, REST APIs, AWS, GCP, Vercel, CloudWatch, Git, Gradle, Jenkins